

Planning plays an important role in traveling. It not only contributes to a trip's success and satisfaction but also to travelers' safety and financial efficiency. However, many travelers may overlook the planning process because they perceive it as complicated and time-consuming. Travel planning involves many aspects, such as destinations, flights, transportation, tourism activities, and budgeting. As a result, planners need to use various tools and platforms to gather information. The objective of this project is to build an iOS mobile application that helps travelers organize trip information and manage travel itineraries more conveniently. The scope of the application focuses on planning features, including creating individual or group trips, adding destinations and activities, scheduling daily itineraries, and tracking estimated expenses.

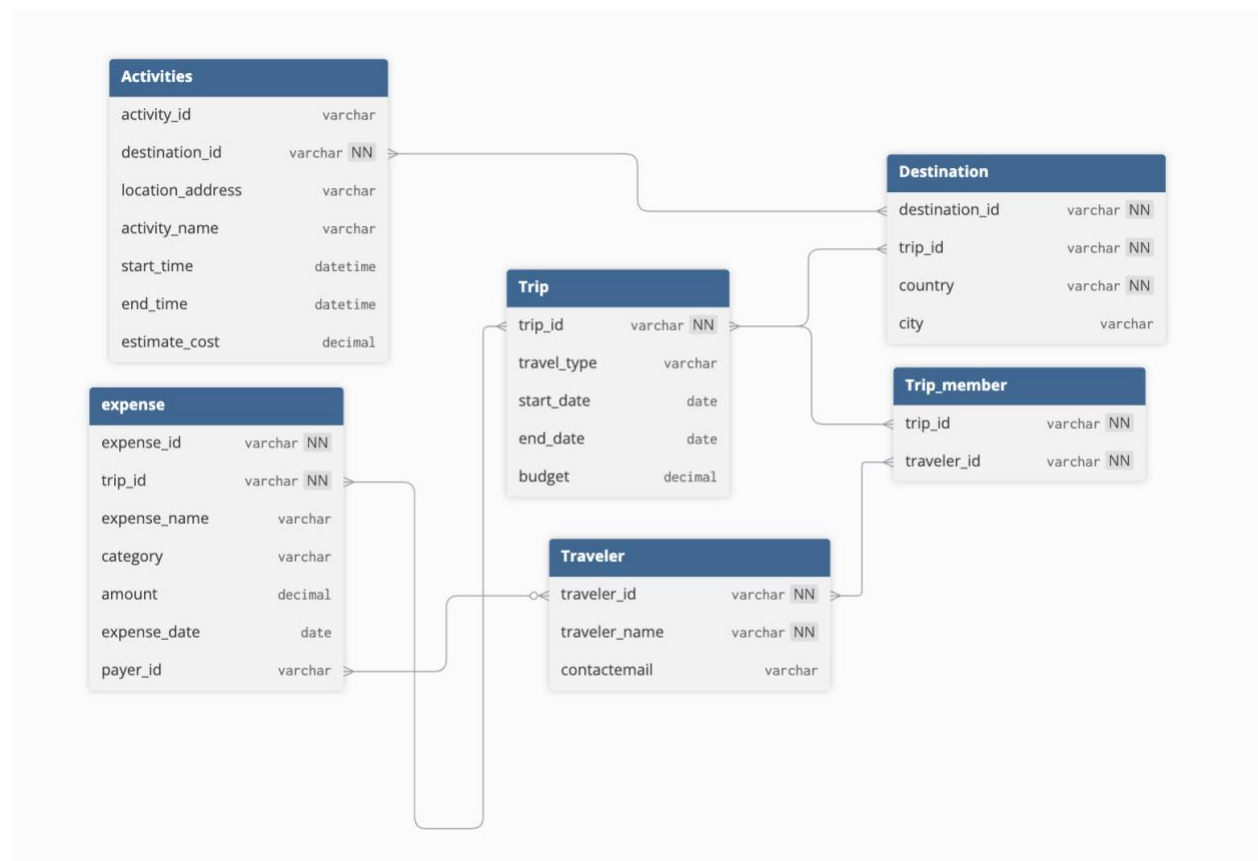


Table	Primary Key	Foreign Key
Trip	trip_id	
Traveler	traveler_id	
Trip_member	trip_id, traveler_id	trip_id, traveler_id
Destination	destination_id	trip_id
Activities	activity_id	destination_id
Expense	Expense_id	trip_id, payer_id

```
CREATE TABLE Trip (  
    trip_id VARCHAR(10) PRIMARY KEY,  
    travel_type VARCHAR(20),  
    start_date DATE,  
    end_date DATE,  
    budget DECIMAL(10,2)  
);
```

```
CREATE TABLE Traveler (  
    traveler_id VARCHAR(10) PRIMARY KEY,  
    traveler_name VARCHAR(100),  
    contact_email VARCHAR(100)  
);
```

```
CREATE TABLE Trip_member (  
    trip_id VARCHAR(10),  
    traveler_id VARCHAR(10),  
  
    PRIMARY KEY (trip_id, traveler_id),  
  
    FOREIGN KEY (trip_id)  
        REFERENCES Trip(trip_id),  
  
    FOREIGN KEY (traveler_id)  
        REFERENCES Traveler(traveler_id)
```

);

```
CREATE TABLE Destination (  
    destination_id VARCHAR(10) PRIMARY KEY,  
    trip_id VARCHAR(10),  
    country VARCHAR(50),  
    city VARCHAR(50),  
  
    FOREIGN KEY (trip_id)  
        REFERENCES Trip(trip_id)  
);
```

```
CREATE TABLE Activities (  
    activity_id VARCHAR(10) PRIMARY KEY,  
    destination_id VARCHAR(10),  
    location_address VARCHAR(255),  
    activity_name VARCHAR(100),  
    start_time DATETIME,  
    end_time DATETIME,  
    estimate_cost DECIMAL(10,2),  
  
    FOREIGN KEY (destination_id)  
        REFERENCES Destination(destination_id)  
);
```

```

CREATE TABLE Expense (
    expense_id VARCHAR(10) PRIMARY KEY,
    trip_id VARCHAR(10),
    expense_name VARCHAR(100),
    category VARCHAR(50),
    amount DECIMAL(10,2),
    expense_date DATE,
    payer_id VARCHAR(10),

    FOREIGN KEY (trip_id)
        REFERENCES Trip(trip_id),

    FOREIGN KEY (payer_id)
        REFERENCES Traveler(traveler_id)
);

```

Example 1: Query listing all the activities in Seoul

$$\pi_{\text{activity_name}} \left(\sigma_{\text{city}='Seoul'} (\text{Activity} \bowtie \text{Destination}) \right)$$

Example 2: Query supporting features displaying expenses in the trip

$$\pi_{\text{amount}} \left(\sigma_{\text{trip}='US1102'} (\text{Expense}) \right)$$

Example 3: Query finding all travelers participating in a group trip

$$\pi_{\text{traveler_id}} \left(\sigma_{\text{trip_id}='US1102'} (\text{Trip_member}) \right)$$

Example 4: Query listing itinerary for a particular date

$$\pi_{\text{activity_name}} \left(\sigma_{\text{start_time} \geq '2026-07-26 00:00:00' \wedge \text{start_time} < '2026-07-26 00:00:00'} (\text{Activity}) \right)$$

Example 5: Query listing listing activities that have not been visited yet

$$\pi_{\text{activity_name}}(\text{Activities}) - \pi_{\text{activity_name}}\left(\sigma_{\text{end_time} < '2026-07-26 12:00:00'}(\text{Activities})\right)$$

I plan to use AI to support the development of the application's back end and database management. Some prompts I plan to use include, for examples :

- What SQL syntax can I use to clean data in my database?
- What concepts and syntax can I use to build a search feature for my application?
- What syntax and logic can I use to automatically generate a travel schedule based on selected activities and budget?